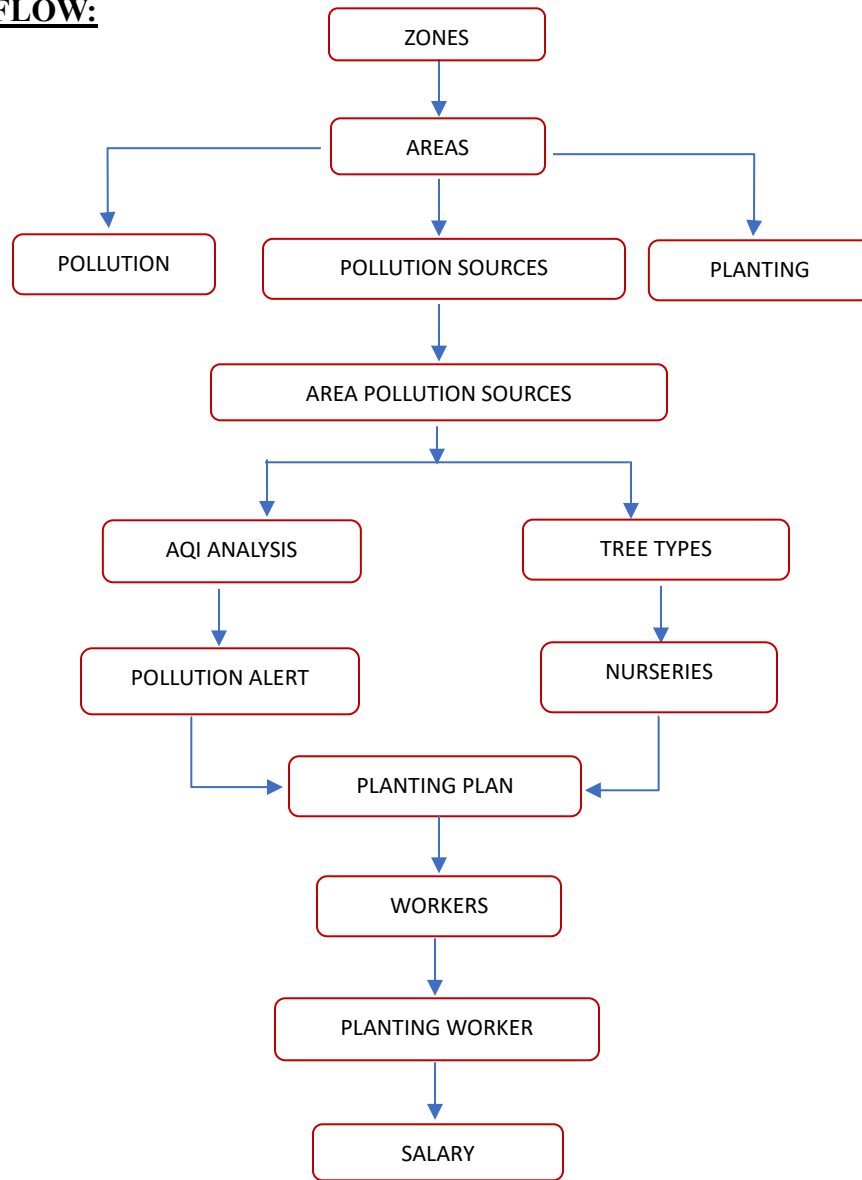


DATABASE FLOW:



BASIC LEVEL:

1. Find the total quantity of trees planted for each area_id.
2. Display all records from the trees table where the tree name starts with 'A'.
3. Display the area name,
4. Find the second-highest AQI from the pollution table.

5. Find the total trees planted in each area, but show only areas where more than 500 trees were planted.
6. Find the area with the highest total funding received.
7. Display areas whose AQI is higher than the average AQI.
8. Find workers who are not assigned to any planting activity.
9. Find the total funding received by each area and display the results from highest funding to lowest.
10. Display the top 3 areas based on total trees planted.
11. Find areas that have planting activity but have not received any funding.
12. Find the minimum salary and maximum salary from the salary table.
13. Find the workers whose salary is equal to the highest salary in the salary table.
14. From the planting table, find the unique tree_name values where the quantity is greater than 50 AND less than 100.
15. Display area_name ,aqi where aqi is between 100 to 300 and also Find the area_id is 2,4and 6.
16. Display the worker_name along with the worker_type in a single column, separated by ' - '.
17. Display the worker_name after removing the first character and the last character from the name.

INTERMEDIATE LEVEL:

1. Display each area name, its AQI, and classify the AQI into an appropriate pollution level using CASE. Group the results by area name and AQI.
2. Rank areas based on the total number of trees planted.
3. Display the areas with total trees and total funding where AQI > 200
4. Find the top 3 areas based on total trees planted using window function.
5. Rank workers by their monthly salary within each worker type, from the highest salary to the lowest salary. Display the worker's name, type, monthly salary, and salary rank.
6. Using LAG(), display each worker's name, monthly salary, and the previous worker's salary, with workers ordered by monthly salary from highest to lowest.

7. Create a view showing each maintenance record, the total maintenance cost of its area, and what percentage of the area's total cost that record represents.
8. Find the total working hours for each worker during the current year.
9. Find the total maintenance cost for each month.
10. Chennai Green Analytics application frequently searches the areas table to find residential areas within a specific zone.

ADVANCED LEVEL:

Stored Procedures

1. Create a stored procedure that accepts **area_id** as an input parameter and returns:
 - area_name
 - AQI
 - total_trees_planted
 - total_funding
2. Create a stored procedure that accepts an **area_id** as input and returns:
 - area_name
 - location_type
 - total_trees_planted

Partition By

3. Write a Query :
 - To partition the planting table by year of planting_date, so that records from 2024, 2025, and 2026 are stored in separate partitions.
 - To retrieve only the 2026 planting records.

Trigger

4. Create a trigger that prevents updating a worker's salary to a negative value.